

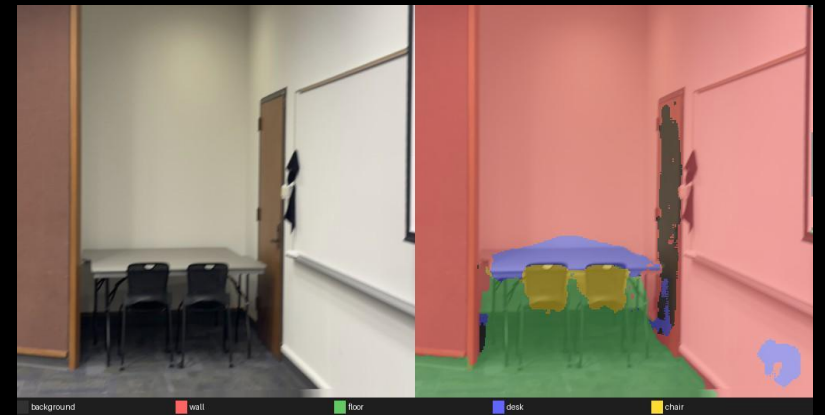
Semantic Segmentation of Indoor Scenes

Problem & Motivation

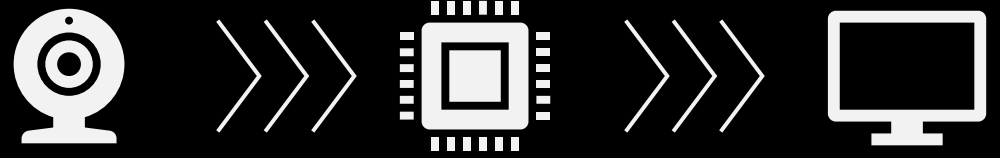
- Our goal is to label every pixel in an image with the use of semantic segmentation
- This will help identify indoor objects such as:
 - Walls
 - Floors/Ceilings
 - Tables
 - Chairs
- The goal is to decompose a complex environment into simple areas which can be used by another system.

-> Approach

- Capture RGB camera input.
- Resize and normalize the frame.
- Run a U-Net segmentation model.
- Overlay the predicted mask in real time.

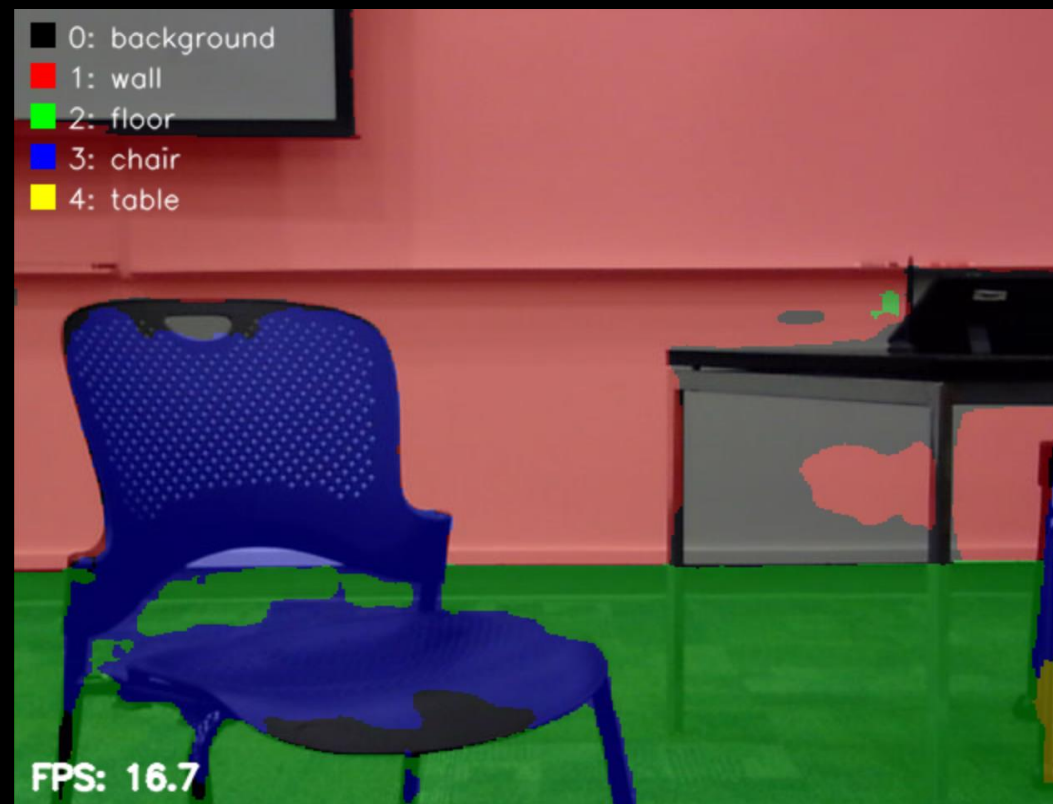
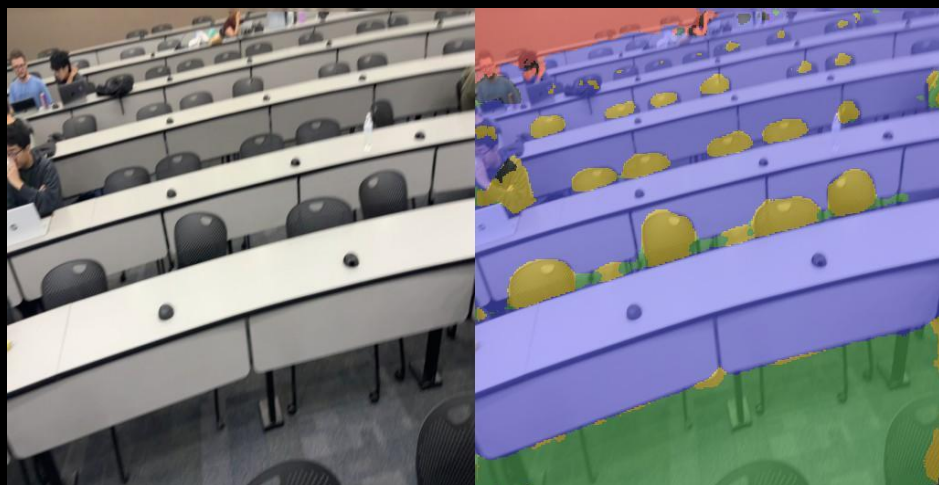


System & Hardware



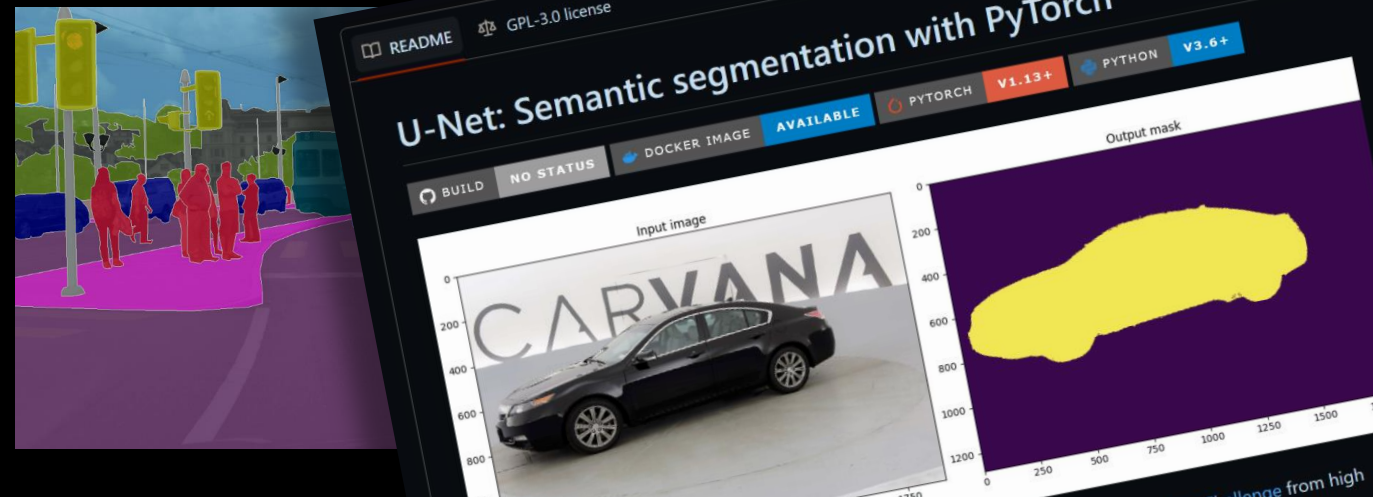
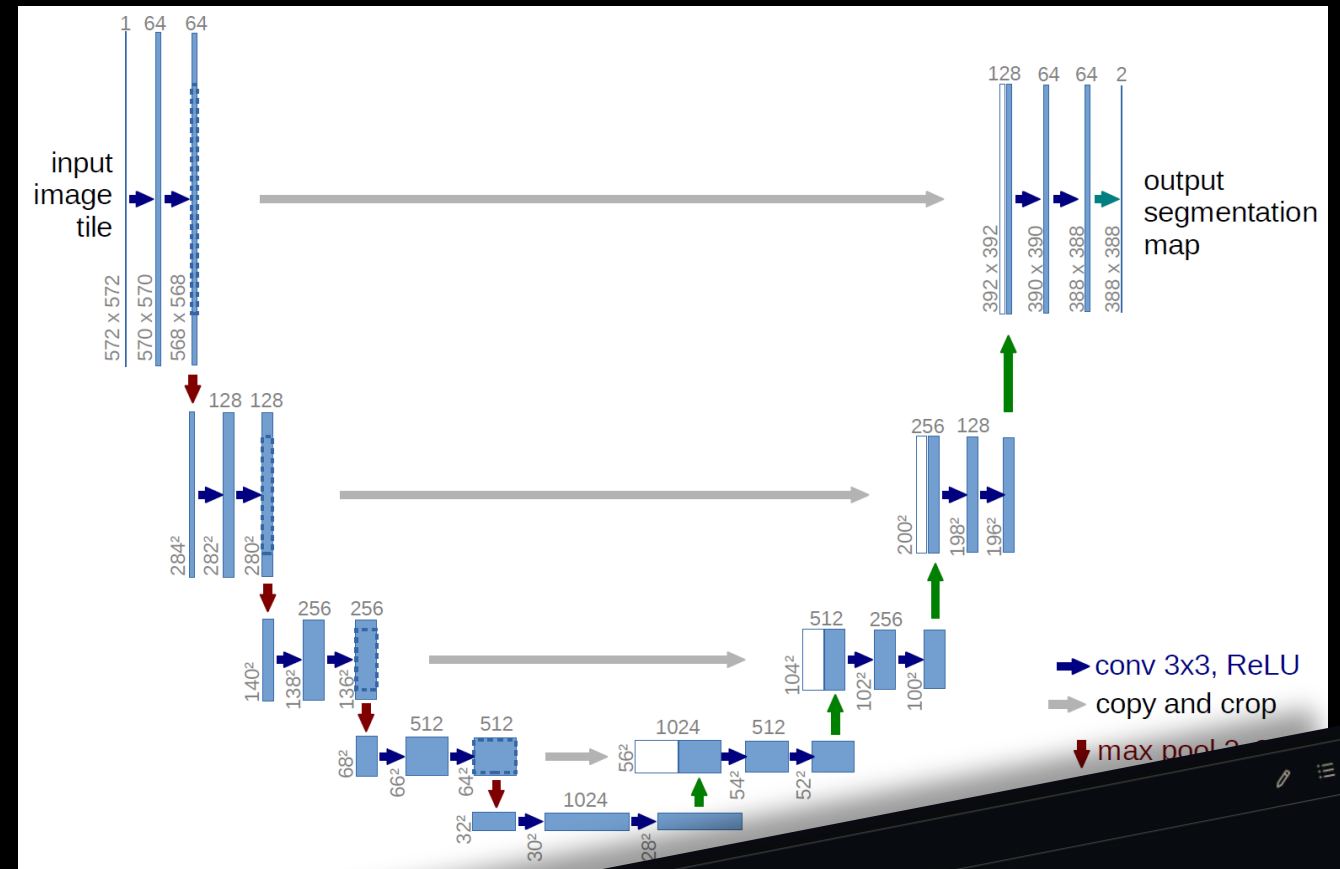
- Jetson Orin Nano, the embedded deployment device
- USB webcam: live RGB input @ 480 × 640
- PyTorch: model training and testing
- ONNX/TensorRT: this is the optimized inference path
- OpenCV: camera capture and visualization
- Target: real-time segmentation at camera native resolution (480 × 640)

System Pipeline



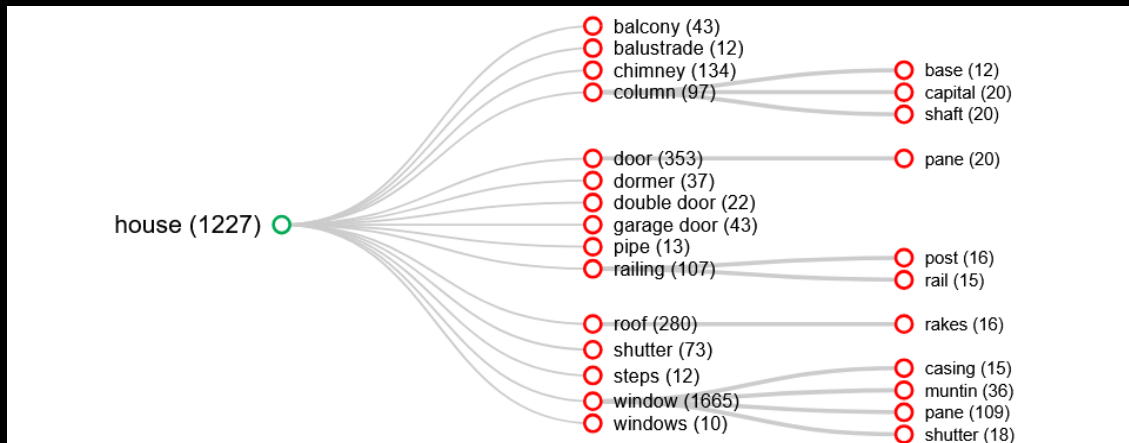
U-Net Architecture

- Encoder learns high-level scene features.
- Decoder upsamples features into a pixel-level mask.
- Skip connections preserve object boundaries.
- Output is a class label for every pixel.



Data & Training

- Indoor images from classrooms and campus were taken.
- Public datasets such as ADE20K were used.
- These images were categorized into interior and outdoor scenes.
 - We used images that had the correct interior classes. (wall, floor, etc)



* this is an example of some of the classes in the ADE20K dataset.



DOWNLOAD

Full Dataset
Register [here](#) to download the ADE20K dataset and annotations. By doing so, you agree to the [terms of use](#).

Toolkit

See our [GitHub Repository](#) for an overview of how to access and explore ADE20K.

Scene Parsing Benchmark

Scene parsing data and part segmentation data derived from ADE20K dataset could be downloaded from [MIT Scene Parsing Benchmark](#).

Terms of Use

See ADE20K's dataset [Terms of Use](#)



Training set
25,574 images
All images are fully annotated with objects and, many of the images have parts too.



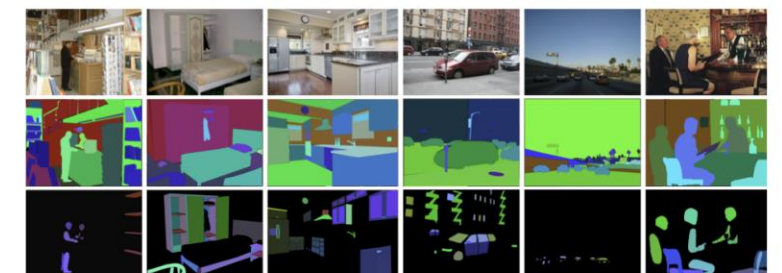
Validation set
2,000 images
Fully annotated with objects and parts.



Test set
Images to be released later.
Consistency set
64 images and annotations used for checking the annotation consistency ([download](#))

OVERVIEW

The annotated images cover the scene categories from the SUN and Places database. Here there are some examples showing the images, object segmentations, and parts segmentations:



The next visualization provides the list of objects and parts and the number of annotated instances. The tree only shows objects with more than 250 annotated instances and parts with more than 10 annotated instances.

Training Overview

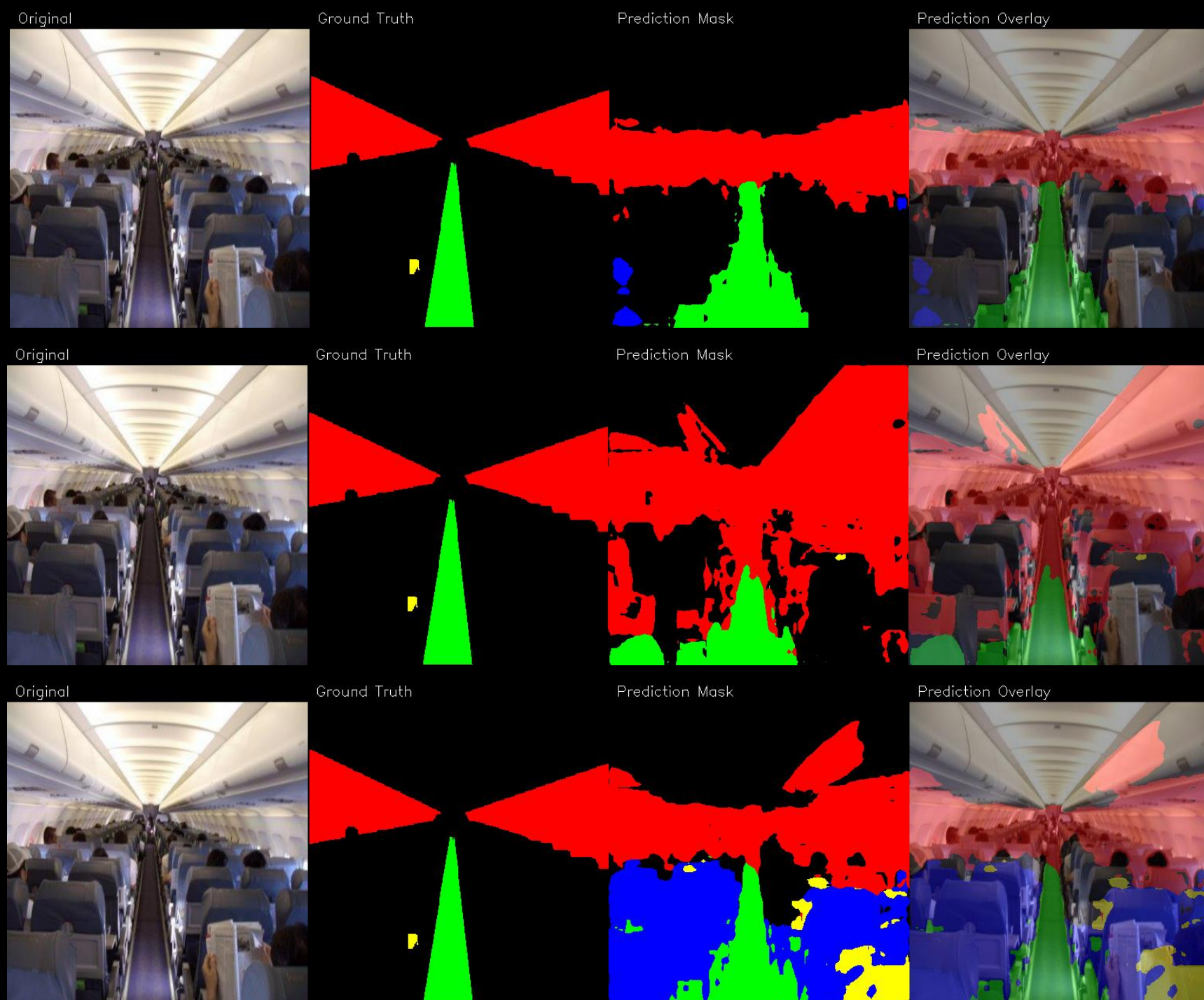
Epoch 1

- Early Epochs introduce a lot of noise that is cleaned up over time.

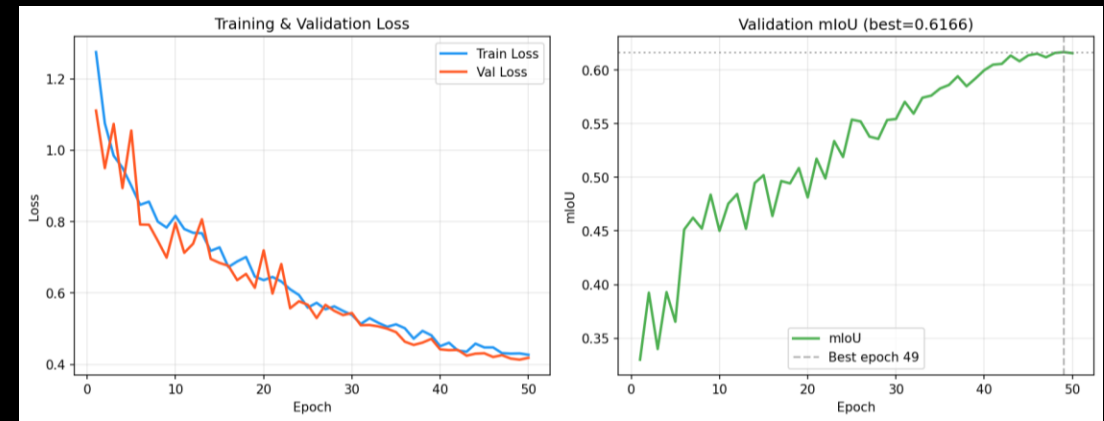
Epoch 2

- Predictions can be unstable.

Epoch n...



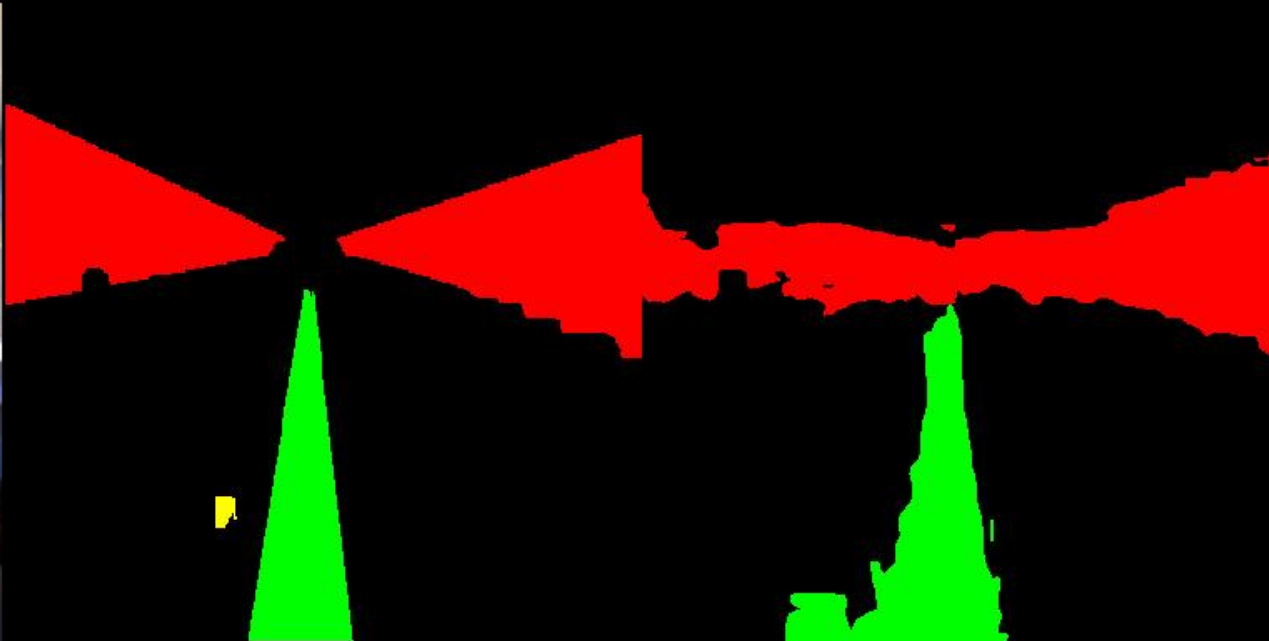
Training Overview



Original



Ground Truth



Prediction Mask

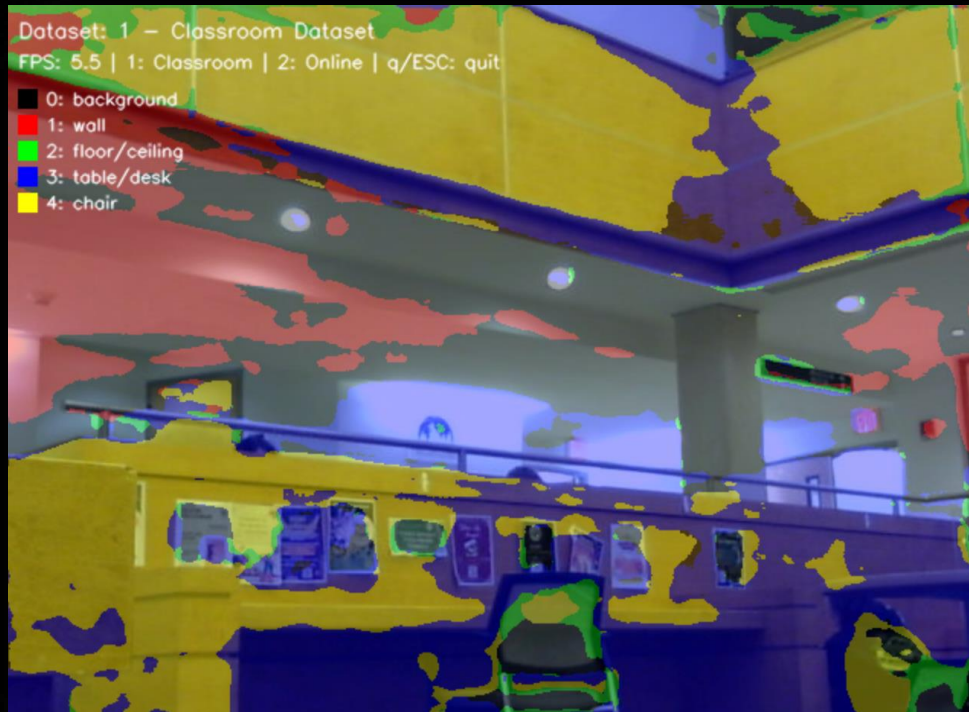


Prediction Overlay

...Epoch ~50!

- By epoch 50, the model better captured large scene regions like walls and floors.

Results/Demo



* model using our (relatively small) dataset.



* model using the online ADE20k dataset, cleaned up to remove outdoor scenes.